

CSA0509 – Database Management Systems

CO1-AT3 TECHNICAL PRESENTATION

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1.ER Diagram

DEFINITION:

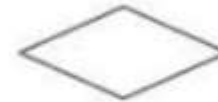
An Entity Relationship (ER) Diagram is a graphical representation of the entities in a database and the relationships between them. It helps in designing the database before implementation.



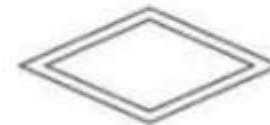
Entity



Weak Entity



Relationship



Weak Relationship



Attribute



Multivalued Attribute

ENTITIES AND ATTRIBUTES

ENTITIES:

- Member
- Book
- Author
- Librarian

ATTRIBUTES

Member

- Member_ID (PK)
- Name
- Email

Book

- Book_ID (PK)
- Title
- Publisher

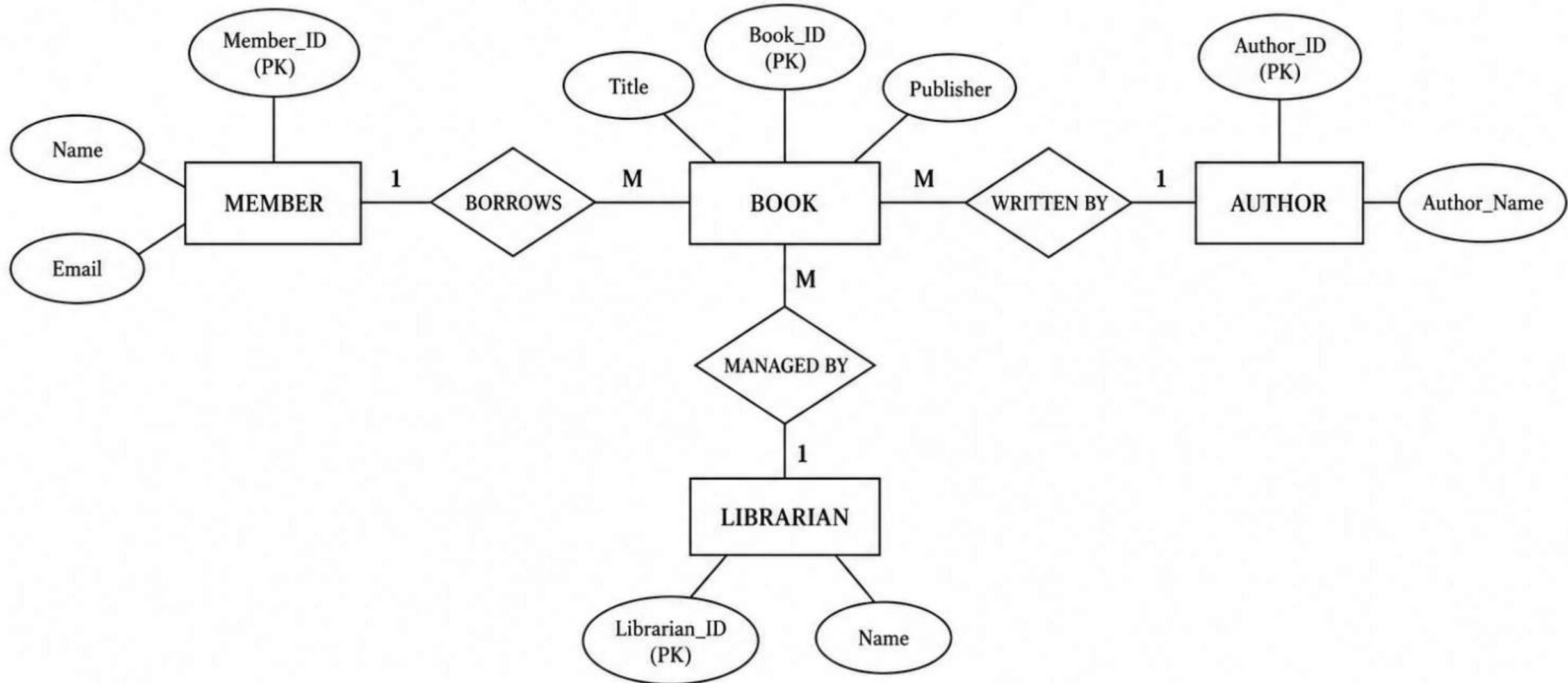
Author

- Author_ID (PK)
- Author_Name

Librarian

- Librarian_ID (PK)
- Name

ER Diagram for an Online Library System



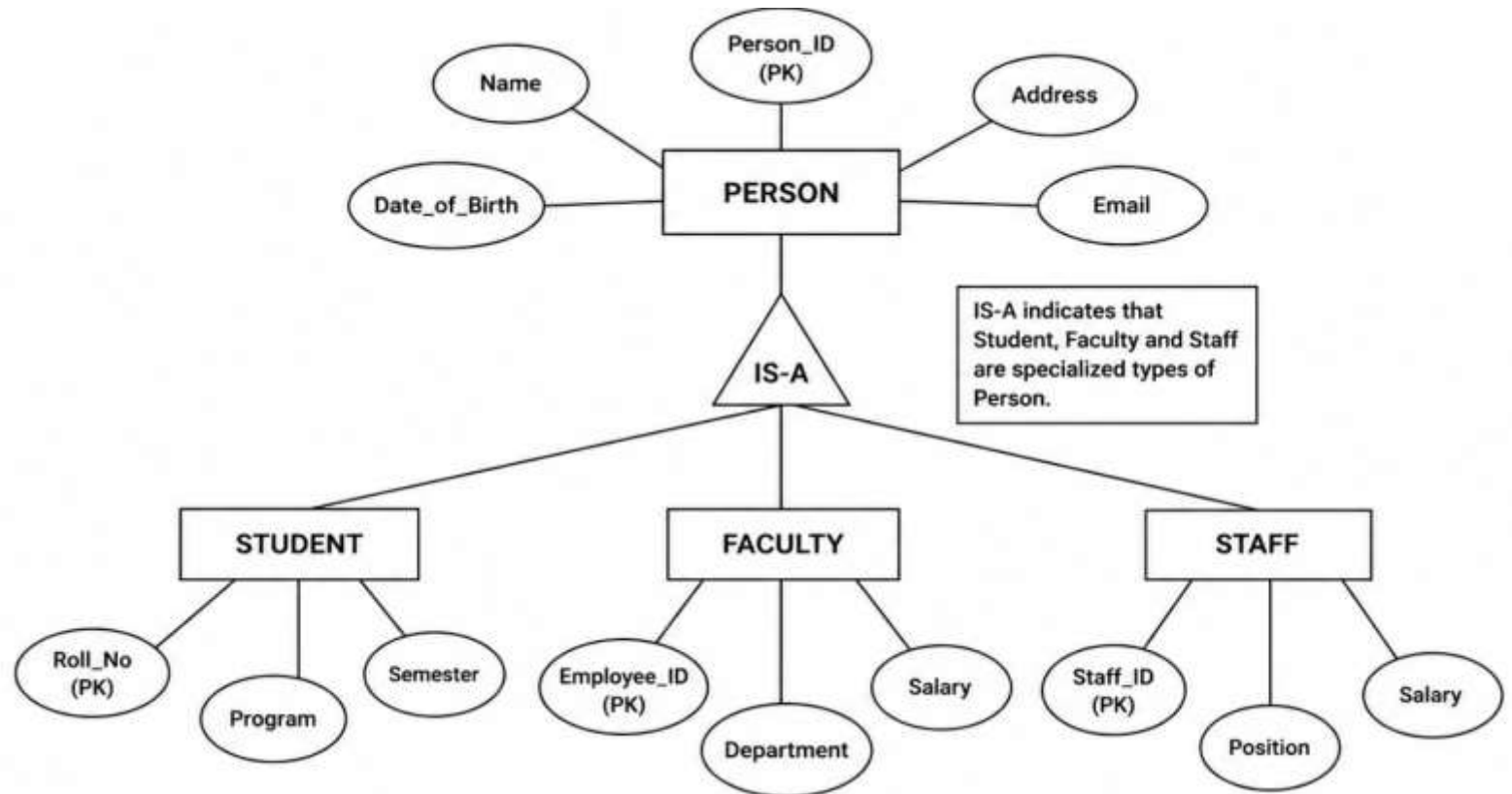
ADVANTAGES

- Eliminates redundancy.
- Maintains data consistency.
- Simplifies database design.
- Makes implementation easier.

2. Enhanced Entity Relationship (EER)

DEFINITION:

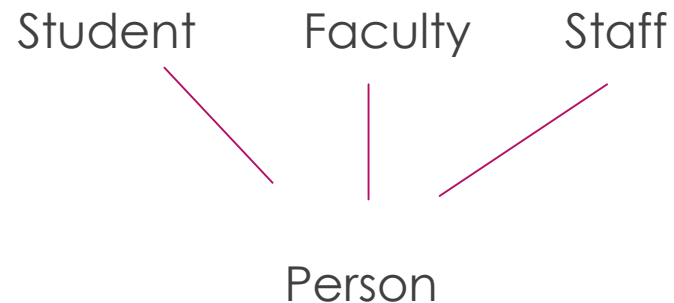
The Enhanced Entity Relationship (EER) Model extends the ER model by introducing inheritance concepts (Generalization and Specialization.)



Generalization

Generalization combines multiple lower-level entities into a higher-level entity.

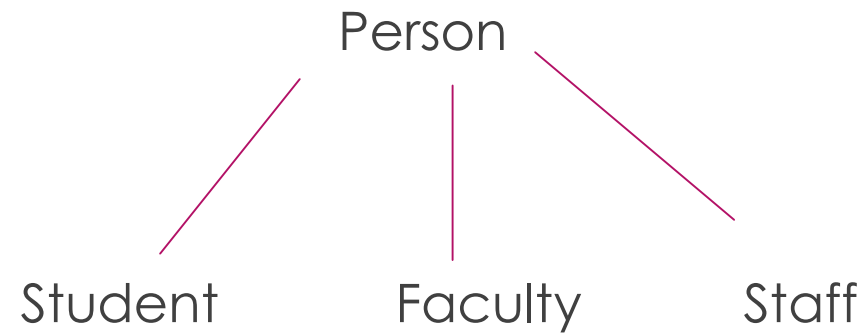
► Example:



Specialization

Specialization divides one higher-level entity into multiple specialized entities.

► Example:



University Case Study

SUPERCLASS

Person

Person_ID

Name

Age

Address

SUBCLASSES

Student

- Roll_No

- Department

- Semester

Faculty

- Employee_ID

- Designation

- Salary

Staff

- Staff_ID

- Position

3. Data Independence

Introduction:

Data Independence is the ability to modify one level of the database without affecting higher levels.

Types of Data Independence

- **Physical Data Independence**
- **Logical Data Independence**

1. Physical Data Independence

Definition:

Changes in physical storage do not affect the logical schema.

Examples:

- Changing indexing methods
- Changing file organization
- Moving data to another disk

2. Logical Data Independence

Definition:

Changes in logical structure do not affect user applications.

Examples:

- Adding new columns
- Creating new tables
- Splitting relations

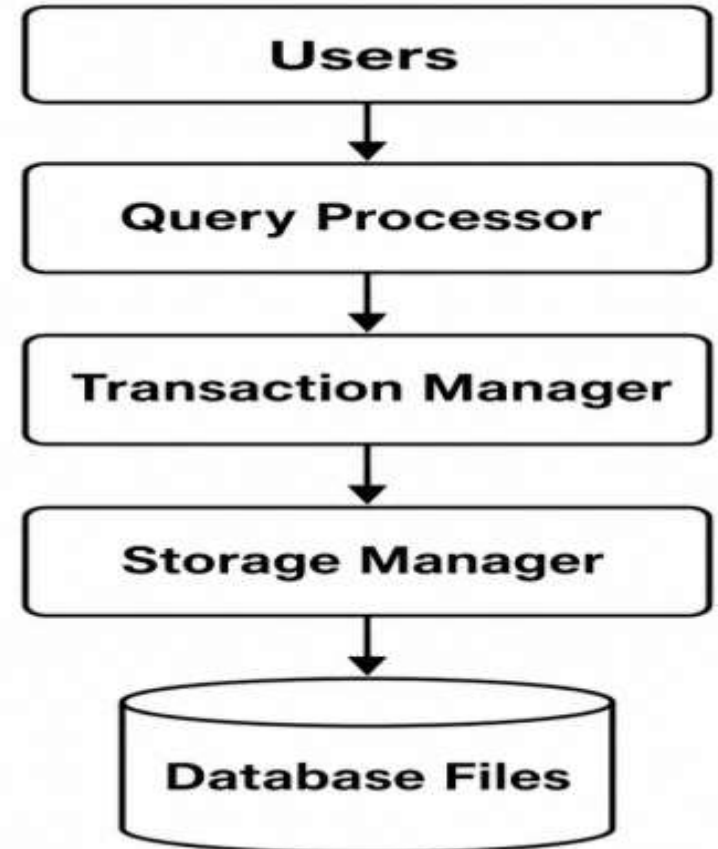
Significance

- ▶ Easier maintenance
- ▶ Better scalability
- ▶ Improved security
- ▶ Reduces application changes
- ▶ Supports future expansion

4. Software Components of a DBMS

INTRODUCTION:

A DBMS consists of software modules that manage storage, processing, and security.



1. Query Processor

Functions:

- Parses SQL queries
- Validates syntax
- Optimizes queries
- Executes SQL statements

Components:

- DDL Interpreter
- DML Compiler
- Query Optimizer
- Query Execution Engine

2. Storage Manager

Functions:

- Stores database files
- Retrieves data
- Manages indexes
- Handles buffers

Components:

- File Manager
- Buffer Manager
- Disk Space Manager

Advantages:

- Efficient storage
- Faster access
- Better disk utilization

3. Transaction Manager

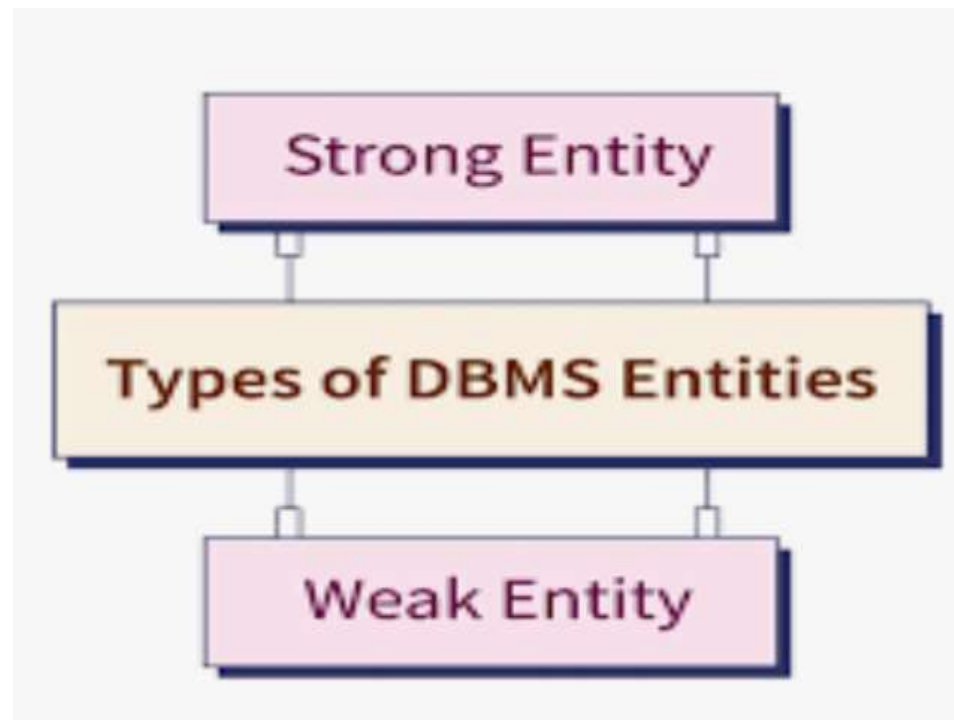
Functions:

- Controls concurrent transactions
- Maintains consistency
- Performs rollback and recovery

Supports ACID Properties

- Atomicity
- Consistency
- Isolation
- Durability

5.TYPES OF ENTITIES:



STRONG ENTITY:

Definition:

A strong entity has its own primary key and exists independently.

EXAMPLE:

Student

- Student_ID (Primary Key)
- Name
- Department

Representation:

Single Rectangle



WEAK ENTITY:

Definition:

A weak entity depends on another entity for its existence.

Example:

Dependent

- Dependent_Name
- Age

It depends on **Employee**.

Representation:

Double Rectangle



REAL-WORLD EXAMPLES

Strong entity:

Student

Student_ID

Name

Department

Weak entity:

Employee

Employee_ID

Name

↓

Dependent

Dependent_Name

Age

Advantages

Strong Entity

- ▶ Independent
- ▶ Easy identification
- ▶ Simple relationships

Weak Entity

- ▶ Reduces redundancy
- ▶ Models dependent objects accurately
- ▶ Maintains referential integrity

**THANK
YOU!**

